

DNS: Domain Name System

A) Using nslookup

1: Basic Domain Lookup

Perform a basic lookup to find the IP address of a given domain name.

Run the following command:

```
nslookup yachaytech.edu.ec
```

Question: What is the IP address of yachaytech.edu.ec?

Expected Output: The command should return the IP addresses associated with the **yachaytech.edu.ec** domain.

2: Reverse Lookup (IP to Domain Name)

Perform a reverse DNS lookup to find the domain name associated with an IP address.

Run the following command:

```
nslookup 8.8.8.8
```

Question: What domain is associated with the IP 8.8.8.8?

Expected Output: It should return a domain name (e.g., dns.google).

3: Query Specific DNS Server

Use a specific DNS server to resolve a domain name.

Run the following command using a custom DNS server:

```
nslookup hpc.cedia.edu.ec 1.1.1.1
```

Question: What is the IP address returned for hpc.cedia.edu.ec when using Cloudflare's DNS server?

Expected Output: IP addresses of hpc.cedia.edu.ec should be returned.

4: Retrieve MX (Mail Exchanger) Records

Use nslookup to find the mail servers associated with a domain.

Run the following command to query the Mail Exchanger (MX) records for yachay.tech.edu.ec:

```
nslookup -type=mx yachaytech.edu.ec
```

Question: What are the mail servers for yachaytech.edu.ec?

Expected Output: The command should return a list of MX records (mail servers) for yachay.tech.edu.ec.

5: Retrieve NS (Name Server) Records

Use nslookup to retrieve the Name Server (NS) records for a domain.

Run the following command to query the name servers for yachay.tech.edu.ec:

```
nslookup -type=ns yachaytech.edu.ec
```

Question: What are the name servers for yachaytech.edu.ec?

Expected Output: The command should return a list of NS records for yachaytech.edu.ec.

6: Query Start of Authority (SOA) Record

Query the Start of Authority (SOA) record of a domain.

Run the following command to get the SOA record for `yachaytech.edu.ec`:

```
nslookup -type=soa yachaytech.edu.ec
```

Question: What information does the SOA record provide about the domain `yachaytech.edu.ec`?

Expected Output: The SOA record includes the primary name server, responsible email address, and other administrative information.

7: Query for Canonical Name (CNAME)

Check if a domain has a CNAME (alias) record.

Run the following command to find the CNAME for www.microsoft.com:

```
nslookup -type=cname www.microsoft.com
```

Question: Does `www.microsoft.com` have a CNAME record? If so, what is it?

Expected Output: The command should return the CNAME record (if any) for `www.microsoft.com`.

8: Debug Mode

Use `nslookup` in debug mode to view detailed output of the DNS lookup process.

Run the following command in debug mode:

```
nslookup -debug yachaytech.edu.ec
```

Question: What additional details are provided in debug mode?

Expected Output: In addition to the regular output, the command will show detailed information like the full query and response.

9: Query a Non-Existent Domain

Investigate how `nslookup` responds to a query for a non-existent domain.

Run the following command:

```
nslookup nonexistentdomain12345.com
```

Question: What is the response when querying a non-existent domain?

Expected Output: An error message stating that the domain does not exist.

B) Using `dnspython`

Repeat the exercises A1 to A9 using “`dnspython`”

LDAP: Setting and Test

1. Installing OpenLDAP

On a Linux-based system (e.g., Ubuntu), use the following commands:

```
sudo apt update
```

```
sudo apt install slapd ldap-utils
```

During installation, it will ask you to set an administrator password. Set a strong password for admin user.

2. Configure LDAP

After installation, reconfigure the LDAP service with the command:

```
sudo dpkg-reconfigure slapd
```

1. Choose the DNS domain name (e.g., example.com).
2. Provide an organization name (e.g., Example Inc.).
3. Select a password for the admin user.
4. Skip database removal and move to the finish of the configuration.

3. Verify LDAP Installation

Check if the LDAP service is running:

```
sudo systemctl status slapd
```

You can use the `ldapsearch` tool to search for entries in the LDAP directory:

```
ldapsearch -x -LLL -b dc=example,dc=com
```

This command should return basic information about your LDAP server.

Adding Entries to the LDAP Directory

4. Create a Basic LDAP Structure:

Let's create an LDIF file (`base.ldif`) to define the organizational structure:

```
dn: ou=People,dc=example,dc=com
```

```
objectClass: organizationalUnit
```

```
ou: People
```

```
dn: ou=Groups,dc=example,dc=com
```

```
objectClass: organizationalUnit
```

```
ou: Groups
```

Load the structure into the LDAP directory:

```
sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f base.ldif
```

5. Adding a User to the Directory:

Create a new user in an LDIF file (`user.ldif`):

```
dn: uid=johndoe,ou=People,dc=example,dc=com
```

```
objectClass: inetOrgPerson
```

```
uid: francisco
```

```
sn: Hidrobo
```

```
cn: Francisco Hidrobo
```

```
userPassword: password
```

Add the user to the directory:

```
sudo ldapadd -x -D cn=admin,dc=example,dc=com -W -f user.ldif
```

Now, you have added a user named Francisco Hidrobo to the LDAP directory.

Accessing the LDAP Directory

6. LDAP Search: Use the `ldapsearch` command to search for the user:

```
ldapsearch -x -LLL -b "dc=example,dc=com" "uid=francisco"
```

This should return the user's information.

7. LDAP Authentication Example:

Using Python, simulate LDAP authentication. First, install the required libraries:

```
pip install python-ldap
```

Here's a simple Python script to authenticate using LDAP:

```
import ldap
ldap_server = "ldap://localhost"
ldap_username = "uid=francisco,ou=People,dc=example,dc=com"
ldap_password = "password"
try:
    conn = ldap.initialize(ldap_server)
    conn.simple_bind_s(ldap_username, ldap_password)
    print("Authentication successful!")
except ldap.INVALID_CREDENTIALS:
    print("Authentication failed!")
except Exception as e:
    print(f"Error: {e}")
finally:
    conn.unbind_s()
```

Modify this script to test different users and passwords.

Special activity

Modify the **Publisher-Subscriber** version of workshop 2 (Part 2) so that the **Publisher** can register a service with an LDAP server and the **Subscriber** can query with the name of the service to obtain the IP address and port to use that service.

Report: You must submit a report (using the same structure as in previous reports) that includes the code (GitHub link), the results (screenshots), and your comments.